

Supp/Deferred Examination

Date : January 2023
Time : 2 hours

Marks available : 71
Full Marks : $66\frac{2}{3}$

*You may answer the questions in any order, provided your solutions are clearly labelled.
Full answers are expected. Please write neatly.
The use of electronic devices is forbidden; this includes calculators.*

1. Read the following theorem and the outline of its proof:

Theorem. If two power series $\sum_{k=0}^{\infty} a_k (z - z_0)^k$ and $\sum_{k=0}^{\infty} b_k (z - z_0)^k$ converge to the same function $f(z)$ on a disc $D_R = \{z \in \mathbb{C} : |z - z_0| < R\}$, where $R > 0$, then $a_k = b_k$ for all integers $k \geq 0$.

Proof. Let n be the smallest integer ≥ 0 such that $a_n \neq b_n$. Then

$$\sum_{k=0}^{\infty} (a_k - b_k)(z - z_0)^k = (z - z_0)^n \sum_{k=0}^{\infty} (a_{n+k} - b_{n+k})(z - z_0)^k, \quad (*)$$

so $\sum_{k=0}^{\infty} (a_{n+k} - b_{n+k})(z - z_0)^k = 0$ for all $z \neq z_0$ in D_R . However, if we put $z = z_0$, then we get that

$$\sum_{k=0}^{\infty} (a_{n+k} - b_{n+k})(z - z_0)^k \neq 0, \quad (**)$$

which leads to a contradiction. $\square$

Now answer the following questions:

- (a) Why can we find a smallest integer $n \geq 0$ such that $a_n \neq b_n$?
- (b) Explain how equation (*) follows.
- (c) Immediately after equation (*), it is claimed that $\sum_{k=0}^{\infty} (a_{n+k} - b_{n+k})(z - z_0)^k = 0$ for all $z \neq z_0$ in D_R . How does this follow?
- (d) Explain why (**) holds if $z = z_0$.
- (e) Why does (**) lead to a contradiction? [2, 3, 3, 1, 3]

2. Find the radius of convergence of each of the following power series:

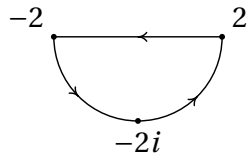
$$(a) \sum_{k=0}^{\infty} \left(\frac{k + (-1)^k}{k} \right)^{(k^2)} (z - i)^k$$

$$(b) \sum_{k=0}^{\infty} \frac{(k!)^2}{(2k)!} (z - i)^k$$

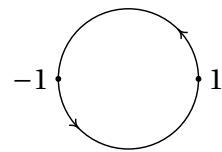
[4, 3]

3. For each of the following, evaluate the integral using the illustrated path γ :

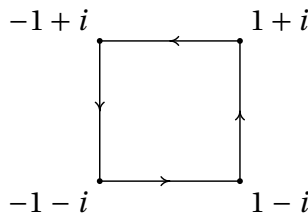
$$(a) \int_{\gamma} |z|^2 dz$$



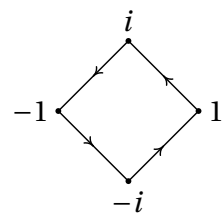
$$(b) \int_{\gamma} \frac{ze^z}{2z+1} dz$$



$$(c) \int_{\gamma} \frac{1}{z \sin(z)} dz$$



$$(d) \int_{\gamma} \frac{z^5}{8z^6-1} dz$$



[6, 4, 5, 5]

4. Use complex variable methods to evaluate

$$(a) \int_0^{\infty} \frac{\cos x}{(9x^2+1)^2} dx$$

$$(b) \text{P.V.} \int_{-\infty}^{\infty} \frac{x^2+2}{x^3+8} dx$$

[8, 11]

5.(a) Find a function f which is analytic on the set $D = \{z \in \mathbb{C} : |z| < 2\}$ and which satisfies

$$f\left(\frac{n-1}{n}\right) = \frac{1}{n+1}$$

for all $n \in \mathbb{N}$.

(b) Explain why there is only one such function f .

[4, 2]

6. How many zeros does the function $p(z) = z^4 - 5z + 3$ have in the annulus $\{z \in \mathbb{C} : 1 < |z| < 2\}$?
[7]